National University of Singapore
School of Computing
CS1101S: Programming Methodology
Semester I, 2026/2027

R9
Environment Model

Problems:

1. Consider the following function declaration:

```
def change(x, new_value):
    x = new_value
```

Consider the following statements:

```
x = 0
change(x, 1)
x
```

What is the value of `x` after the above statements are evaluated? Why? Explain your answer using the environment model.

2. Consider the following program:

```
x = 0
def make_adder(x):
    return lambda y: x + y
make_adder(1)(x)
```

What is the result of evaluating the program? Why? Explain your answer using the environment model.

3. Consider the following program, which uses continuation-passing style to append two llists:

```
def app(current_xs, ys, c):
    if is_none(current_xs):
        return c(ys)
    else:
        return app(tail(current_xs), ys,
                   lambda x: c(pair(head(current_xs), x)))

def append_iter(xs, ys):
    return app(xs, ys, lambda x: x)

append_iter(llist(1, 2, 3), llist(4, 5))
```

The longer the first llist, the more memory the evaluation of this program consumes. Where is this memory consumption represented in the environment model, and where in the e-stepper?